CUSTOMER SHOPPING BEHAVIOR ANALYSIS

A data-driven review of purchasing patterns, customer segments and subscription behavior

3,900 Purchase records	**18** Dataset columns	**37** Missing review ratings

1. Project overview

This project examines customer shopping behavior using 3,900 purchase records across multiple product categories. The analysis explores spending patterns, customer segments, product preferences and subscription behavior, with the aim of informing business decisions.

2. Analysis objectives

- Compare revenue across customer groups, including gender and age group.
- Examine discount usage, shipping choices and subscription status.
- Identify highly rated products and frequently purchased products by category.
- Segment customers using purchase history and examine repeat-purchase behavior.

Scope note
The report presents descriptive analysis of the supplied transaction dataset. The results should be interpreted as patterns within these records, not as proof of causation or a guarantee of future customer behavior.

3. Dataset & preparation

Dataset summary

Records: 3,900 **Columns:** 18 **Missing values:** 37 in Review Rating.

Feature groups

• **Customer demographics:** age, gender, location and subscription status.
• **Purchase details:** item purchased, category, purchase amount, season, size and color.
• **Shopping behavior:** discount applied, promo code used, previous purchases, purchase frequency, review rating and shipping type.

Data preparation in Python

• Loaded the dataset with pandas and reviewed its structure using `df.info()` and `df.describe()`.
• Checked missing values and imputed missing Review Rating values using the median rating within each product category.
• Standardized column names to snake_case.
• Created `age_group` by binning customer ages and `purchase_frequency_days` from purchase data.
• Checked whether `discount_applied` and `promo_code_used` were redundant; the report states that `promo_code_used` was dropped.
• Connected the Python workflow to PostgreSQL and loaded the cleaned DataFrame for SQL analysis.

SQL analysis questions

The PostgreSQL analysis covers: revenue by gender; high-spending discount users; top five products by rating; average purchase amount by shipping type; subscribers versus non-subscribers; discount-dependent products; customer segmentation; top three products per category; repeat buyers and subscription status; and revenue by age group.

Data integrity
The figures and query-result tables on the following pages are retained from the supplied report. No values, rankings, labels or calculated results have been changed. Wording has been edited for clarity only.

4. SQL results | Revenue, discount users & product ratings

The source tables show revenue by gender, customers who used discounts while spending above the average purchase amount, and the five highest-rated products.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	89

Total rows: 839 Query complete 00:00:(

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

4. SQL results | Shipping & subscription comparison

The source tables compare average purchase amounts by shipping type and summarize average spend and total revenue by subscription status.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

4. SQL results | Discount dependence & customer segments

The source tables list products with the highest percentage of discounted purchases and the customer counts in each purchase-history segment.

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

4. SQL results | Category leaders, repeat buyers & age groups

The source tables show the top three products in each category, repeat-buyer subscription counts, and revenue by age group.

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

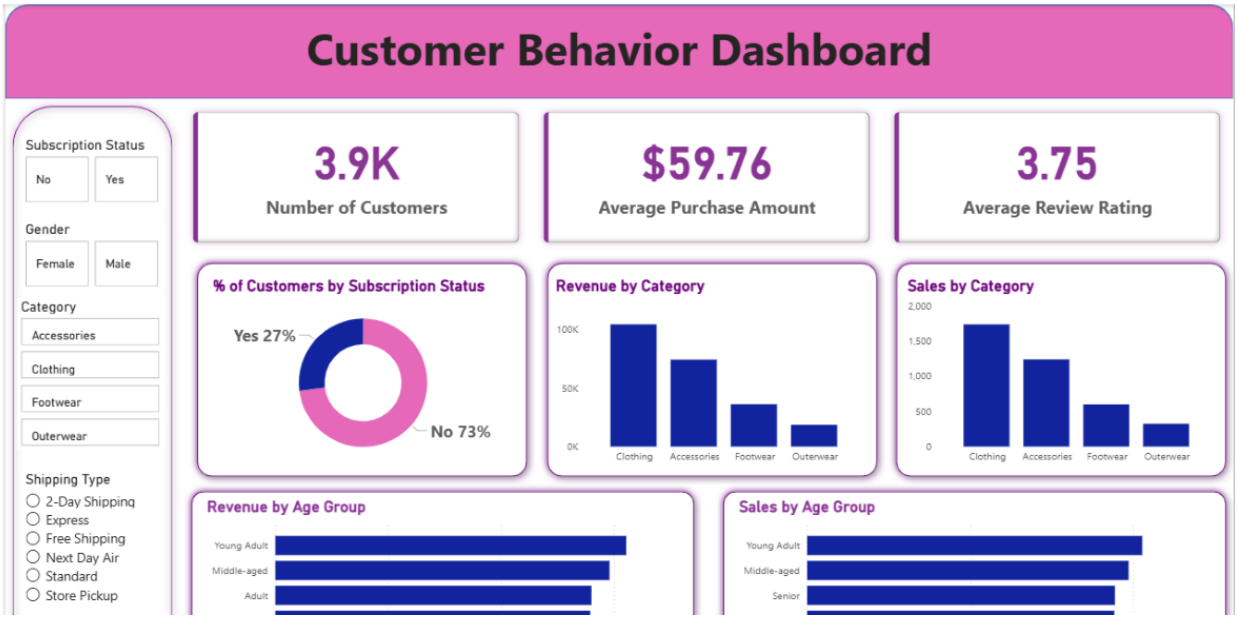
5. Power BI dashboard

The original dashboard visual is retained as supplied. Its displayed metrics and chart values have not been recalculated or altered.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business recommendations

The following actions are aligned with the analysis topics and recommendations in the original report. They are strategic suggestions, not measured outcomes from the dataset.

1. Boost subscriptions
Promote the benefits available to subscribers and communicate the value of subscription clearly.

2. Strengthen customer loyalty
Consider rewards for repeat buyers to encourage continued engagement and support movement into the “Loyal” segment.

3. Review discount policy
Evaluate discount effectiveness alongside profitability and margin impact before expanding discount offers.

4. Improve product positioning
Feature highly rated and best-selling products in relevant campaigns, while keeping ratings and sales performance as distinct measures.

5. Use targeted marketing
Use the age-group revenue analysis to inform audience targeting. The original report also proposes focusing on express-shipping users; validate this segment's value before increasing spend.

Interpretation reminder
The dataset contains 3,900 purchase records. Unless the underlying data and query logic establish otherwise, record-level results should not automatically be described as unique-customer behavior. Likewise, observed associations (such as repeat purchasing and subscription status) do not establish that one causes the other.

End of report